

1.1 System Overview

DocuLens AI consists of five major architectural layers: (1) Frontend, (2) API/Application Layer, (3) Document Intelligence Layer, (4) Retrieval and Generation Layer, and (5) Infrastructure Layer. The system is architected for strict evidence traceability, deterministic citation validation, and high-performance multimodal retrieval.

1.2 Document Intelligence Components

The Document Intelligence Layer uses PyMuPDF for PDF parsing, page geometry analysis, and page image rendering. Text chunking uses semantic structure boundaries with configurable chunk size and overlap.

1.3 Phase 4 · Hybrid Retrieval Fusion

The hybrid retrieval step combines Dense Retrieval (BGE embeddings), BM25 (sparse lexical), and Visual Retrieval (CLIP vision embeddings) using either Reciprocal Rank Fusion (RRF) or normalized weighted score fusion.

Figure 1: Multimodal Ingestion and Retrieval Pipeline Diagram

Ingestion flow: Upload PDF -> PyMuPDF Page Parsing -> Dense Chunker + Visual Page Renderer (150 DPI) -> Qdrant Vector Collections (document_chunks & visual_pages) + In-Memory BM25 Index. Query flow executes parallel dense/sparse/visual search, applies RRF fusion, and executes cross-encoder reranking.

5. Document Intelligence & Traceability Standards

For complete evidence traceability across the pipeline, required metadata fields must include where applicable: `document_id`, `page_number`, `chunk_id`, and source/evidence reference. Chunks without valid page associations are rejected during validation.

6. Quality Gates and Grounded Generation

Deterministic citation validation ensures strict grounding against retrieved evidence. The AnswerGenerator inspects all evidence tags (e.g. [Evidence 1]) and verifies that referenced page numbers match actual retrieved chunks. Answers with missing evidence or hallucinated citations are flagged as ungrounded.

7. Rate Limiting and Telemetry

In-process fixed-window rate limiting is enforced per client IP with a default window of 60.0 seconds and maximum 60 requests. System health metrics track pipeline latency percentiles (p50, p95) and citation verification rates.